

YOGESH A

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[Mail](#) [Linked in](#) [GitHub](#) [PortFolio](#)

OBJECTIVE

Results- Oriented **pre-final year** student with a strong foundation in Data Science, Machine Learning, and AI. Demonstrated experience building complex models (Vision Transformers), and Insight-driven ML dashboards. Skilled in **Python-based automation and Generative AI**, with proven ability to deliver actionable insights. Seeking an internship to apply robust analytical skills to new challenges in ML, DL, and Intelligent systems.

EDUCATION

B.Tech Artificial Intelligence and Data Science

Saranathan College of Engineering - Autonomous

Tiruchirappalli, Tamil Nadu

September 2023 - Present

- **Cumulative GPA: 7.56/10**
- **Relevant Coursework:** Machine Learning, Deep Learning, Artificial Intelligence, Fundamentals of Data Science, OOP Concepts, Data Exploration and Visualization, Computer Networks, Data Structures and Algorithms.

SKILLS

Languages	: C, Python, Java ,JavaScript, SQL .
Frameworks	: Django, FastAPI.
Tools & Technologies	: Power BI, MySQL, Git, GitHub, TensorFlow, Scikit-learn, LangChain, LangGraph, RAG.
Soft Skills	: Adaptability, Problem-Solving, Leadership, Teamwork, Time Management.

PROJECTS

ExcelAI – Voice Activated GenAI Spreadsheet Assistant

- **Built a Full-Stack AI Assistant** using **Django** and **Gemini 2.0** for voice-driven Google Sheets automation.
- **Engineered Agentic Workflows** with **LangGraph** to execute autonomous row updates and cell highlighting.
- **Integrated Google Sheets API** for seamless, real-time cloud synchronization and bi-directional data flow.
- **Developed a CSV Import System** for instant bulk data syncing with an auto-refreshing live preview.
- **Technologies:** Python, gemini-3-flash-preview, Django, LangChain, Langgraph, GoogleSheet API, JavaScript.
- **Github:** [Link](#) .

PostgreSLM-Domain Specific AI Assistant for PostgreSQL Administration

- Developed a Full-Stack AI powered PostgreSQL Assistant by integrating a fine-tuned **Mistral-7B** Small language model(SLM) with a **Hybrid Retrieval-Augmented Generation (RAG) pipeline** for accurate ,context-aware database support.
- Fine-tuned **Mistral-7B-Instruct** using Unsloth(QLoRA/PEFT) on PostgreSQL domain-specific datasets ,significantly improving SQL generation ,query Understanding and technical response quality.
- Designed a **Hybrid Retrieval Engine** using **Langchain Ensemble Retriever** ,combining BM25 keyword search with **FAISS** semantic retrieval to minimize hallucinations and enhance factual accuracy.
- **Technologies:** Python, Mistral-7B, QLoRA, PEFT, Langchain, FAISS, BM25, FastAPI, React.js
- **Github:** [Link](#) .

Medical Image Classification using Vision Transformers

- Developed and trained an advanced deep learning model for a college mini-project focused in Medical image classification.
- Implemented a Vision Transformer (ViT) architecture to achieve high accuracy in classifying medical scans, showcasing expertise in modern deep learning and computer vision.
- **Technologies:** Django, Python, DeepLearning ,PyTorch ,Vision Transformers, Scikit-learn.

LANGUAGE PROFICIENCY

English - Full Professional Proficiency

Tamil - Native Proficiency